ROCCO ZHANG

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EDUCATION

Princeton University, B.S.E. in Computer Science, Minor in Operations Research GPA: 3.92 | May 2028
Courses: Distributed Sys., Probability and Stochastic Sys., ML, Computer Vision, Programming Sys.

Thomas Jefferson High School for Science & Technology GPA: 4.7/4.0 (Top 1%)

EXPERIENCE

Copperlane AI (YC W26) – *Software Engineering Intern* May 2026 – Aug 2026

- Designed and shipped to production an end-to-end LLM pipeline that turns lender guideline documents into runnable, cited mortgage eligibility products, leveraging specialized LLM stages with deterministic validation, RAG-backed Q&A, and an AI-assisted admin review page.
- Re-architected core data models and loan search from on-demand data API calls into asynchronous polling and a projected Postgres search directory, equipped with fuzzy matching, query classification, paginated responses, and JOIN-less reads.
- Shipped full-stack features on a FastAPI + React platform, including AI-assistant chat reliability, Stripe billing, outbound safety gates, durable long-running workflows, and fixes to production errors like ReDoS vulnerabilities.

University of Maryland – *Research Intern* May 2025 – Aug 2025

- Researched a new natural language processing (NLP) embedding method under Prof. Zhou for sentence embeddings using mixture of experts and proved with enhanced performance than baselines.

Sanghani Center for AI – *Research Intern* May 2023 – Oct 2023

- Developed an advanced NLP algorithm with multi-task learning to classify mental health conditions including bipolar from Reddit, achieving ~90% accuracy. Presented paper at IEEE conference in Italy.

PROJECTS

- **The Forum** (Princeton TigerApps) **–** Developed a web-based campus event hub and recommender to improve event visibility using Express, email scrapers, and multi-signal recommendations with vectorized databases.
- **Saynario –** Developed a full-stack (FastAPI + WebSockets/React) voice AI language learning app, with a low-latency STT-LLM-TTS streaming pipeline. Designed an adaptive AI coach with custom scenario generation based on persisted user profiles. Won Best Use of ElevenLabs at HackPrinceton 26'.
- **URSecretary –** Developed a RAG agentic secretary, with web search, call making, and Email handling, with per-user personalization and secrets storage on a full-stack app (Express + Postgres/React).
- **Multi-agent Robotics** (Senior Research) – Developed a multi-agent robotic solution for search-and-rescue, training customized object detection with graph neural network-coordinated inter-agent communication.
- **FTC Robotics** – Designed, built and programmed robots for the FTC competition, wrote and optimized computer vision-assisted autonomous and finite state machine driver control programs in Java.

PUBLICATIONS

- Zhang, R., Sarkar, S., Alhamadani, A., & Lu, C.-T. "More than Just a Diagnosis: A Multi-Task Approach to Analyzing Bipolar Disorder on Reddit via DeMHeM" *IEEE International Conference on Big Data* 2023
- "How do Computers See? A Picture Book Intro to Computer Vision", author and illustrator

AWARDS

- HackPrinceton 2026 Winner
- USAPhO Semifinalist | USACO Gold | US Presidential Scholars Semifinalist
- Junior Science and Humanities Symposium 3rd place oral presenter
- FTC Robotics State Finalist & Inspire Award
- Eagle Scout

SKILLS

- Python (PyTorch, FastAPI, Pandas), Java, C/C++, Typescript/Node.js, Go, React, PostgreSQL, AWS